# TrES-3b

**R-0360**

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-3_250609` · created 2026-07-18T13:04:19+00:00

**BENCHMARK: ACCURATE — fit consistent with published values**

## Results

| | |
|---|---|
| **Mid-Transit Time (Tmid)** | 2460835.7047 +/- 0.0016 BJD_TDB |
| **Ratio of Planet to Stellar Radius (Rp/R*)** | 0.179 +/- 0.013 |
| **Transit depth (Rp/Rs)^2** | 3.21 +/- 0.46 [%] |
| **Orbital Inclination (inc)** | 82.75 +/- 0.51 |
| **Airmass coefficient 1 (a1)** | 1.018 +/- 0.012 |
| **Airmass coefficient 2 (a2)** | -0.014 +/- 0.012 |
| **Scatter in the residuals of the lightcurve fit is** | 1.21 % |
| **Best Comparison Star** | #1 - [451, 346] |
| **Optimal Method** | PSF photometry |
| **Transit Duration (day)** | 0.0638 +/- 0.0039 |

## Accuracy vs published ephemeris (ExoClock/NEA)

| | |
|---|---|
| **Rp/R■ fit vs published** | 0.179 ± 0.013 vs 0.16309 (deviation 1.22σ) |
| **Duration fit vs published** | 0.0638 d vs 0.0567 d (deviation 1.83σ) |
| **Mid-time O–C** | -2.0 min |
| **Depth SNR** | 13.8 |
| **Residual scatter** | 1.21 % |

## Light curve

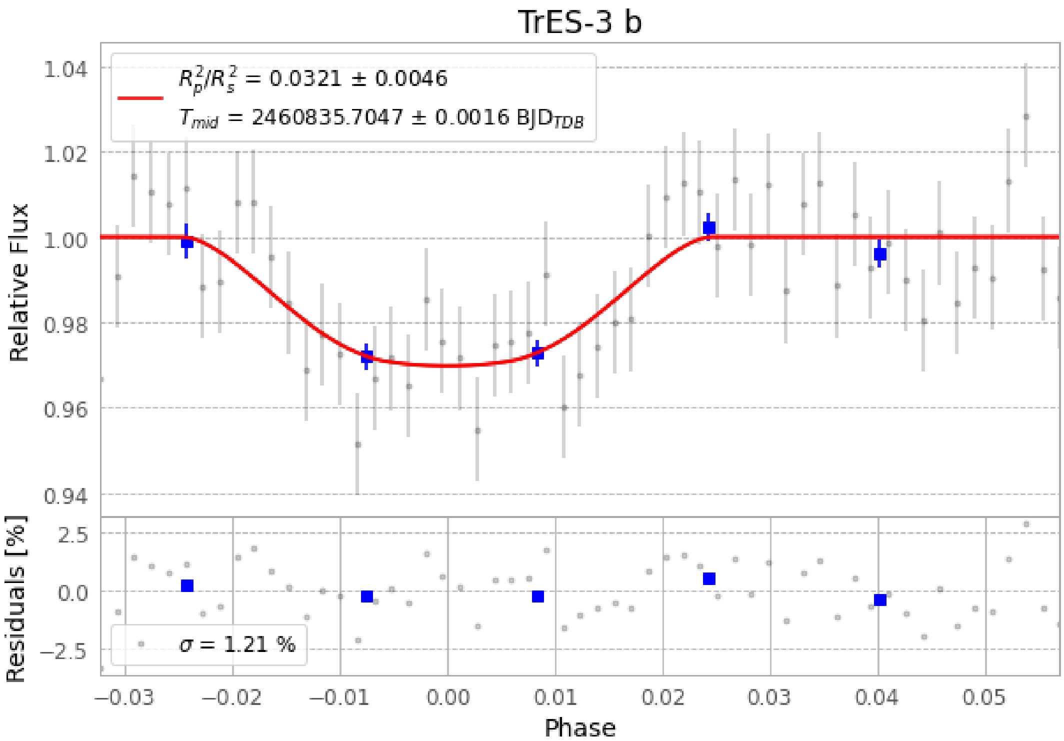


Figure 1 — FinalLightCurve_TrES-3 b_2025-06-08.png (SHA-256 23bca06117015beb…)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [345, 187], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 132e780ea4b6a143…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 9e597cf`

## Data provenance

57 FITS frames (37 MB), `TRES-3250609034816.FITS` → `TRES-3250609063615.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-250609.log (a202e917e18f…), FinalLightCurve_TrES-3 b_2025-06-08.png (23bca0611701…), FinalLightCurve_TrES-3 b_2025-06-08.csv (1dd716bf3614…)

## Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)